

Experiment 15 — Iris Flower Classification using Naive Bayes

AIM

To implement Iris Flower Classification using the Naive Bayes classification algorithm in Python.

SOFTWARE USED

- Python
- Google Colab / Jupyter Notebook
- NumPy
- Scikit-learn

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required Python libraries.
3. Load the built-in Iris dataset.
4. Separate the input features and target classes.
5. Divide the dataset into training and testing data.
6. Create a Gaussian Naive Bayes classifier.
7. Train the classifier using the training dataset.
8. Predict the Iris flower classes using the test dataset.
9. Calculate the classification accuracy.
10. Display the confusion matrix.
11. Display the predicted and actual classes.

CODE

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()

# Input features
X = iris.data

# Target classes
y = iris.target

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Gaussian Naive Bayes model
model = GaussianNB()

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Display confusion matrix
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
# Display actual and predicted classes
print("\nActual Classes:")
print(y_test)
print("\nPredicted Classes:")
print(y_pred)
import matplotlib.pyplot as plt
import seaborn as sns
# Create confusion matrix
cm = confusion_matrix(y_test, y_pred)
# Plot confusion matrix
plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Naive Bayes")
plt.show()
```

OUTPUT

Accuracy: 1.0

Confusion Matrix:

```
[[10 0 0]
```

```
 [ 0 9 0]
```

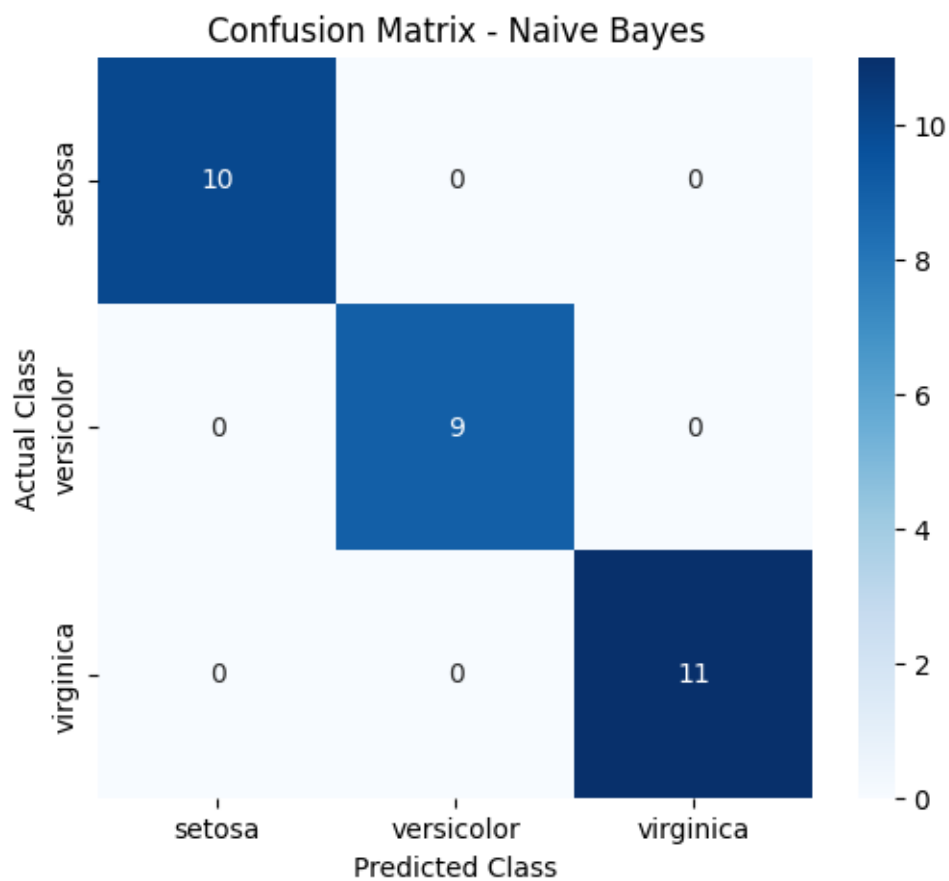
```
 [ 0 0 11]]
```

Actual Classes:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```

Predicted Classes:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```



RESULT

Thus, the Iris Flower Classification was successfully implemented using the Gaussian Naive Bayes classifier, and the Iris flowers were classified into their respective species.